

Soft Drink Bottling Experiment Example 5.3

The GLM Procedure

Class Level Information		
Class	Levels	Values
PRESSURE	2	25 30
SPEED	2	200 250
CARBONATION	3	10 12 14
REPLICATE	2	1 2

Number of Observations Read	24
Number of Observations Used	24

Soft Drink Bottling Experiment Example 5.3

The GLM Procedure

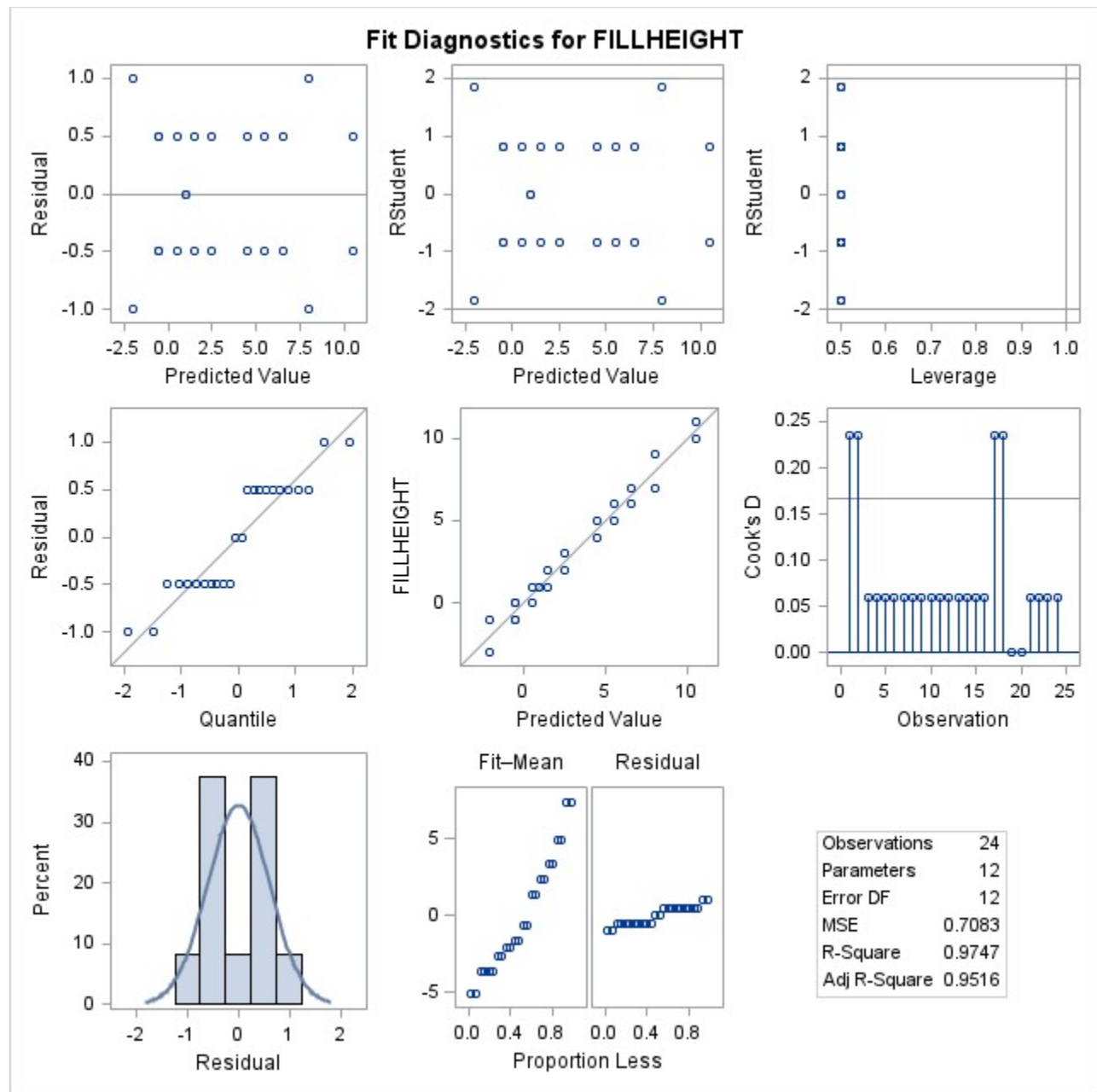
Dependent Variable: FILLHEIGHT

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	11	328.1250000	29.8295455	42.11	<.0001
Error	12	8.5000000	0.7083333		
Corrected Total	23	336.6250000			

R-Square	Coeff Var	Root MSE	FILLHEIGHT Mean
0.974749	26.93201	0.841625	3.125000

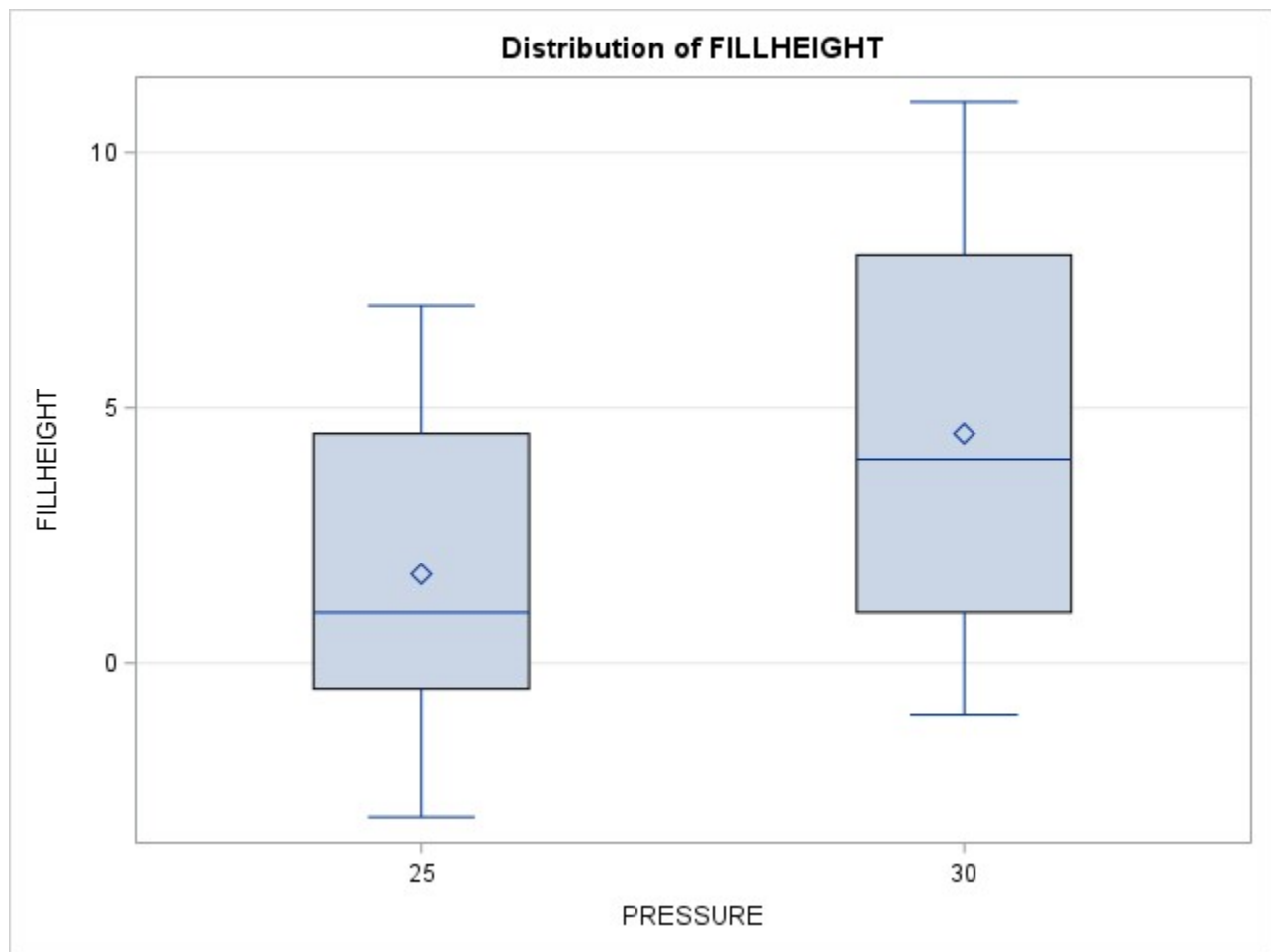
Source	DF	Type I SS	Mean Square	F Value	Pr > F
PRESSURE	1	45.3750000	45.3750000	64.06	<.0001
SPEED	1	22.0416667	22.0416667	31.12	0.0001
PRESSURE*SPEED	1	1.0416667	1.0416667	1.47	0.2486
CARBONATION	2	252.7500000	126.3750000	178.41	<.0001
PRESSURE*CARBONATION	2	5.2500000	2.6250000	3.71	0.0558
SPEED*CARBONATION	2	0.5833333	0.2916667	0.41	0.6715
PRESSU*SPEED*CARBONA	2	1.0833333	0.5416667	0.76	0.4869

Source	DF	Type III SS	Mean Square	F Value	Pr > F
PRESSURE	1	45.3750000	45.3750000	64.06	<.0001
SPEED	1	22.0416667	22.0416667	31.12	0.0001
PRESSURE*SPEED	1	1.0416667	1.0416667	1.47	0.2486
CARBONATION	2	252.7500000	126.3750000	178.41	<.0001
PRESSURE*CARBONATION	2	5.2500000	2.6250000	3.71	0.0558
SPEED*CARBONATION	2	0.5833333	0.2916667	0.41	0.6715
PRESSU*SPEED*CARBONA	2	1.0833333	0.5416667	0.76	0.4869



Soft Drink Bottling Experiment Example 5.3

The GLM Procedure



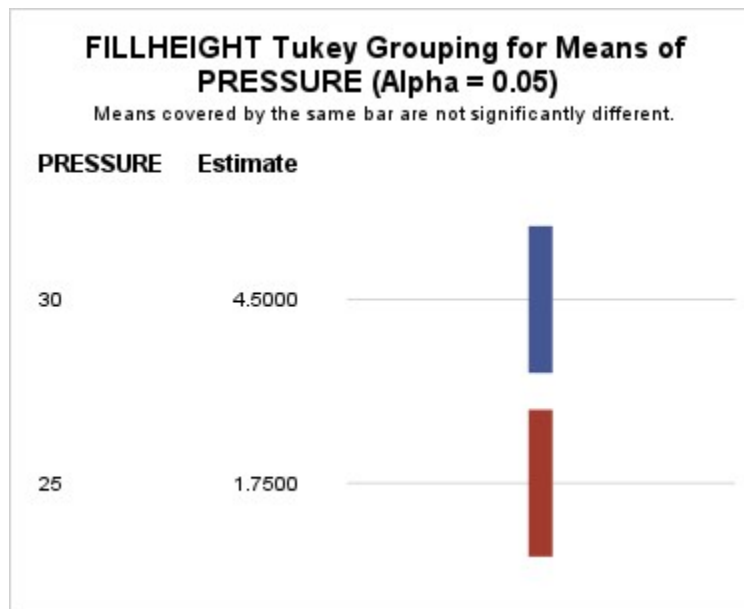
Soft Drink Bottling Experiment Example 5.3

The GLM Procedure

Tukey's Studentized Range (HSD) Test for FILLHEIGHT

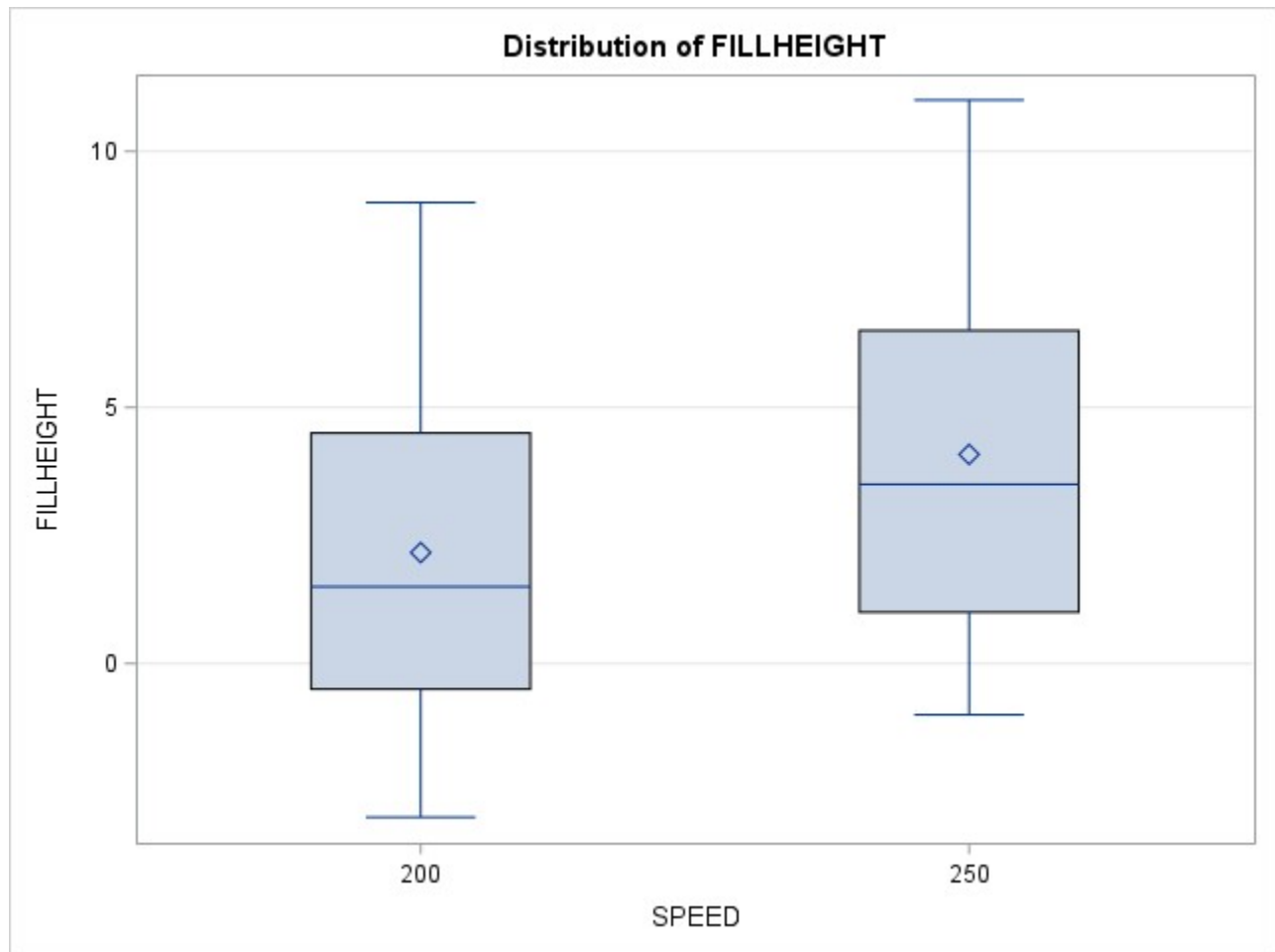
Note: This test controls the Type I experimentwise error rate, but it generally has a higher Type II error rate than REGWQ.

Alpha	0.05
Error Degrees of Freedom	12
Error Mean Square	0.708333
Critical Value of Studentized Range	3.08118
Minimum Significant Difference	0.7486



Soft Drink Bottling Experiment Example 5.3

The GLM Procedure



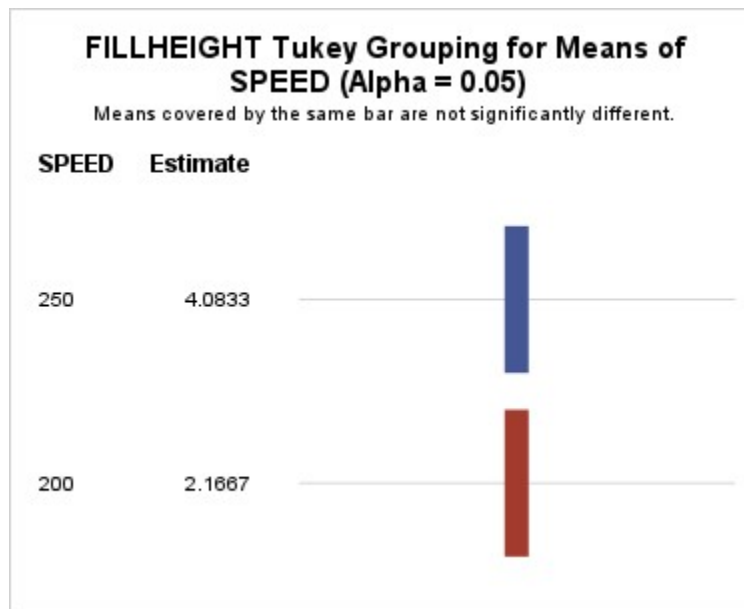
Soft Drink Bottling Experiment Example 5.3

The GLM Procedure

Tukey's Studentized Range (HSD) Test for FILLHEIGHT

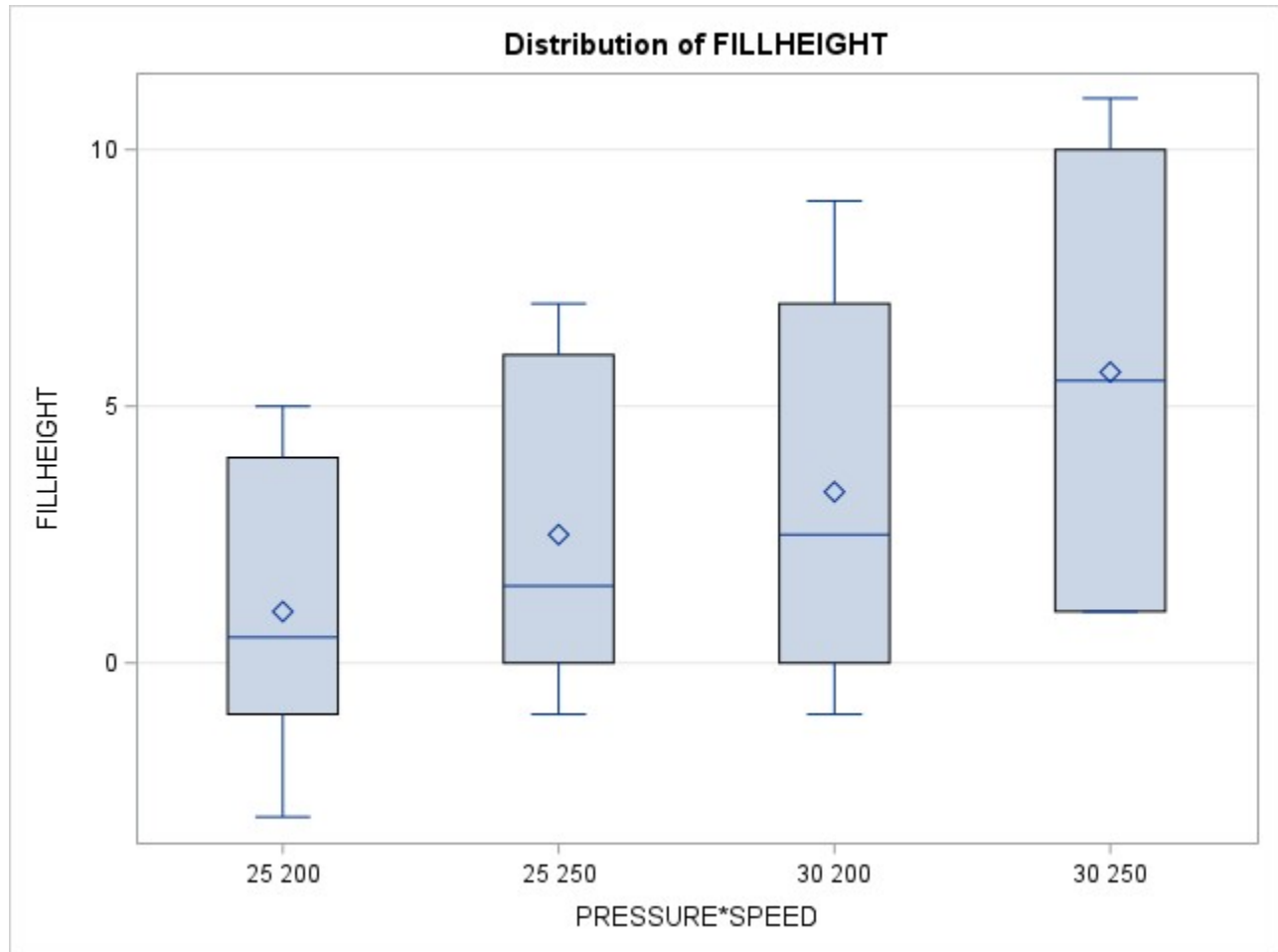
Note: This test controls the Type I experimentwise error rate, but it generally has a higher Type II error rate than REGWQ.

Alpha	0.05
Error Degrees of Freedom	12
Error Mean Square	0.708333
Critical Value of Studentized Range	3.08118
Minimum Significant Difference	0.7486

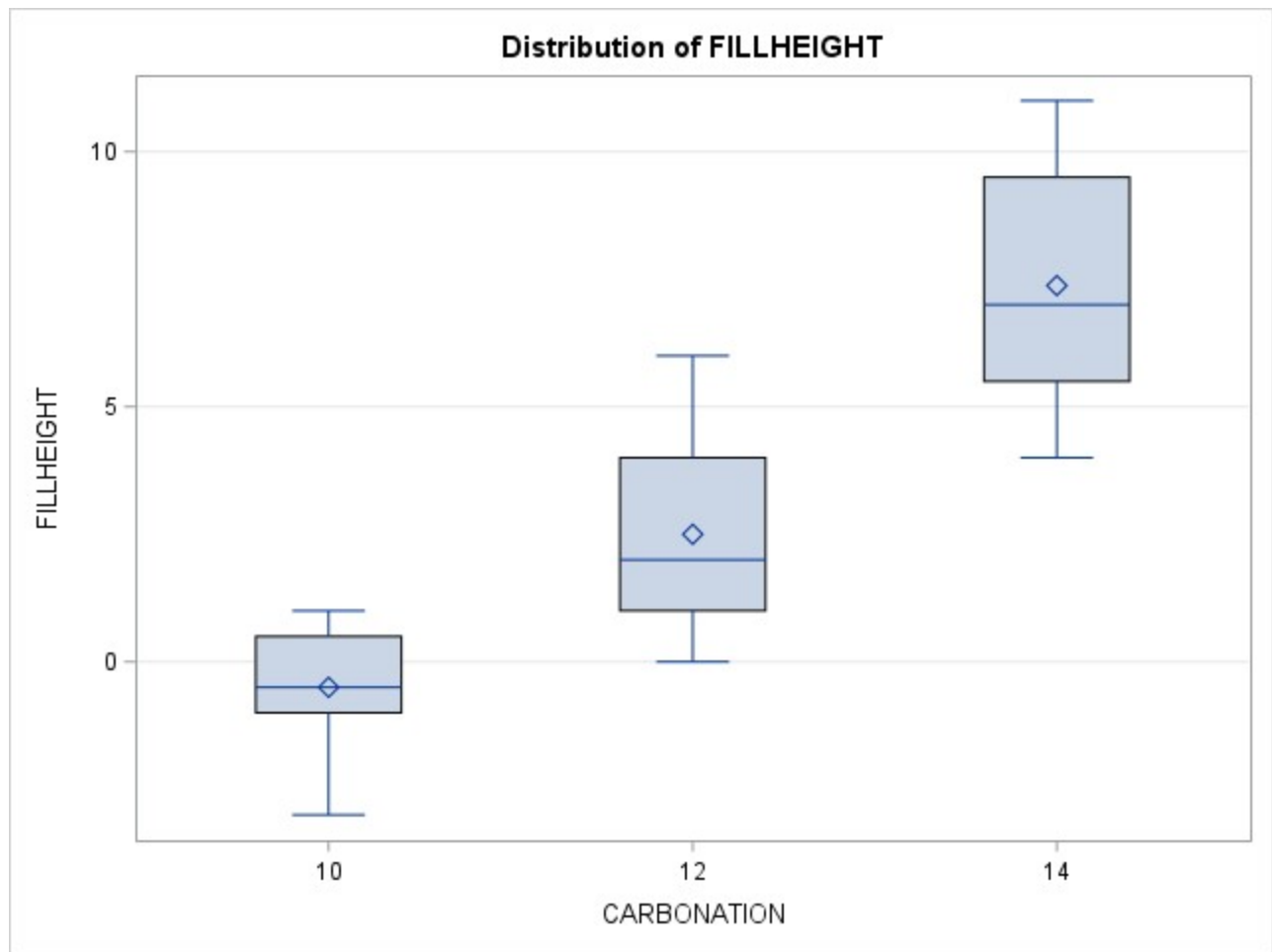


Soft Drink Bottling Experiment Example 5.3

The GLM Procedure



Level of PRESSURE	Level of SPEED	N	FILLHEIGHT	
			Mean	Std Dev
25	200	6	1.00000000	3.03315018
25	250	6	2.50000000	3.27108545
30	200	6	3.33333333	3.93276832
30	250	6	5.66666667	4.27395211



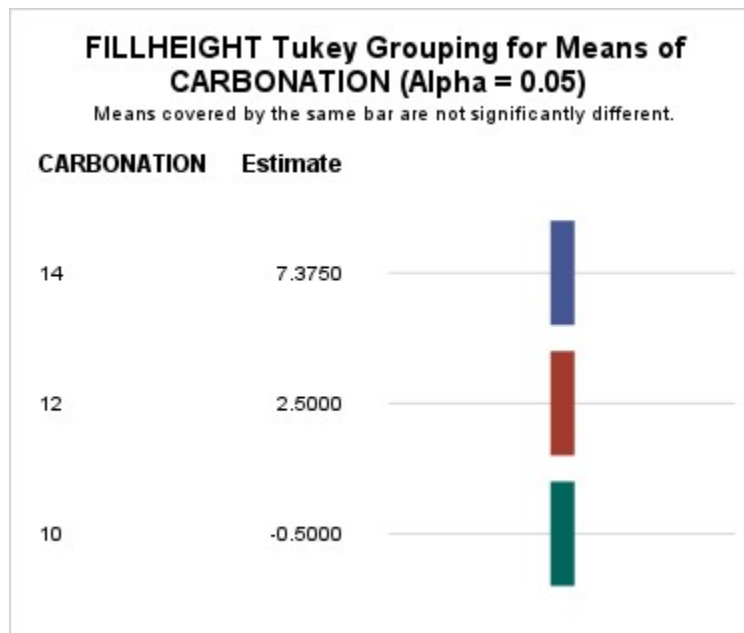
Soft Drink Bottling Experiment Example 5.3

The GLM Procedure

Tukey's Studentized Range (HSD) Test for FILLHEIGHT

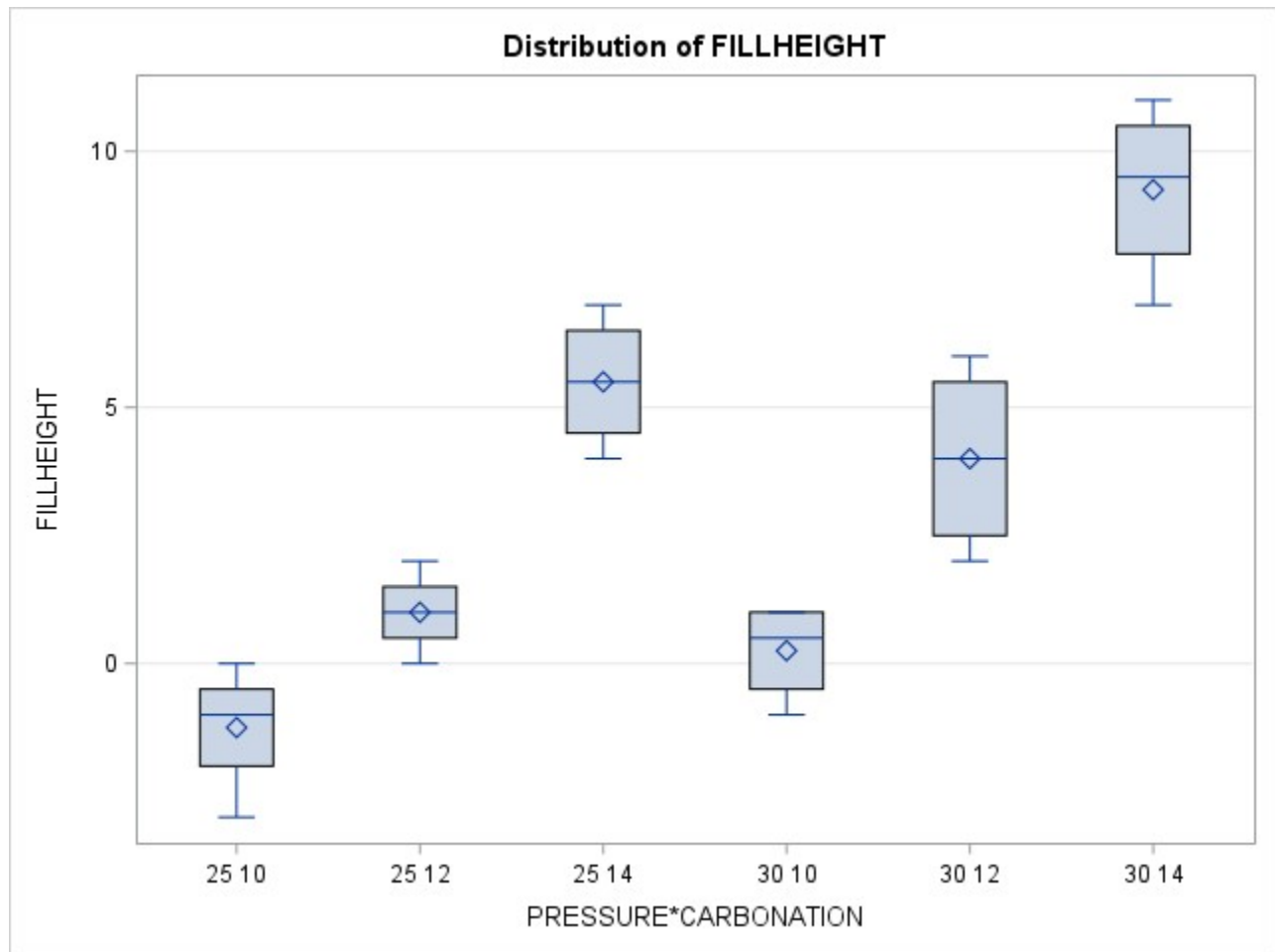
Note: This test controls the Type I experimentwise error rate, but it generally has a higher Type II error rate than REGWQ.

Alpha	0.05
Error Degrees of Freedom	12
Error Mean Square	0.708333
Critical Value of Studentized Range	3.77289
Minimum Significant Difference	1.1227

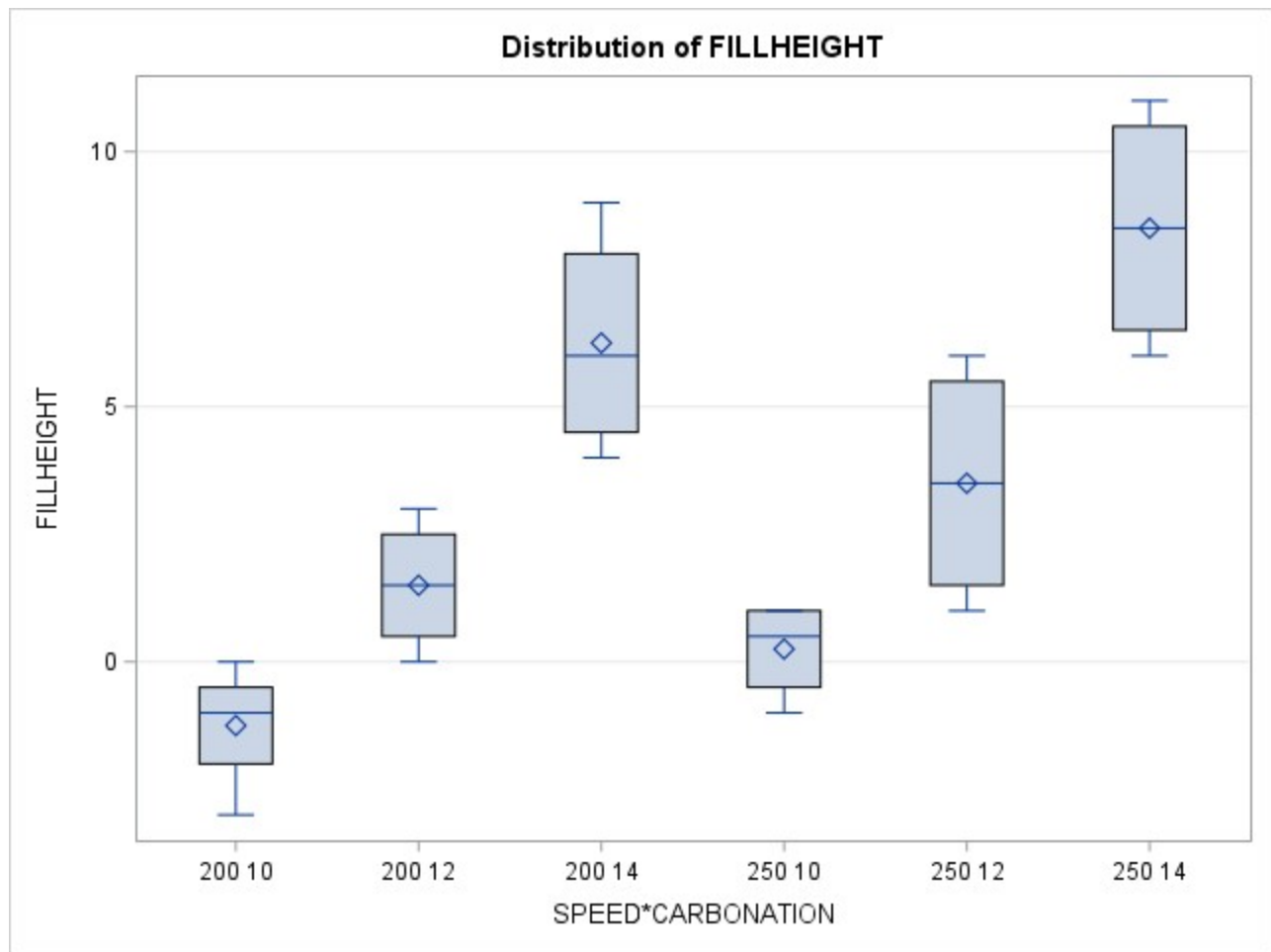


Soft Drink Bottling Experiment Example 5.3

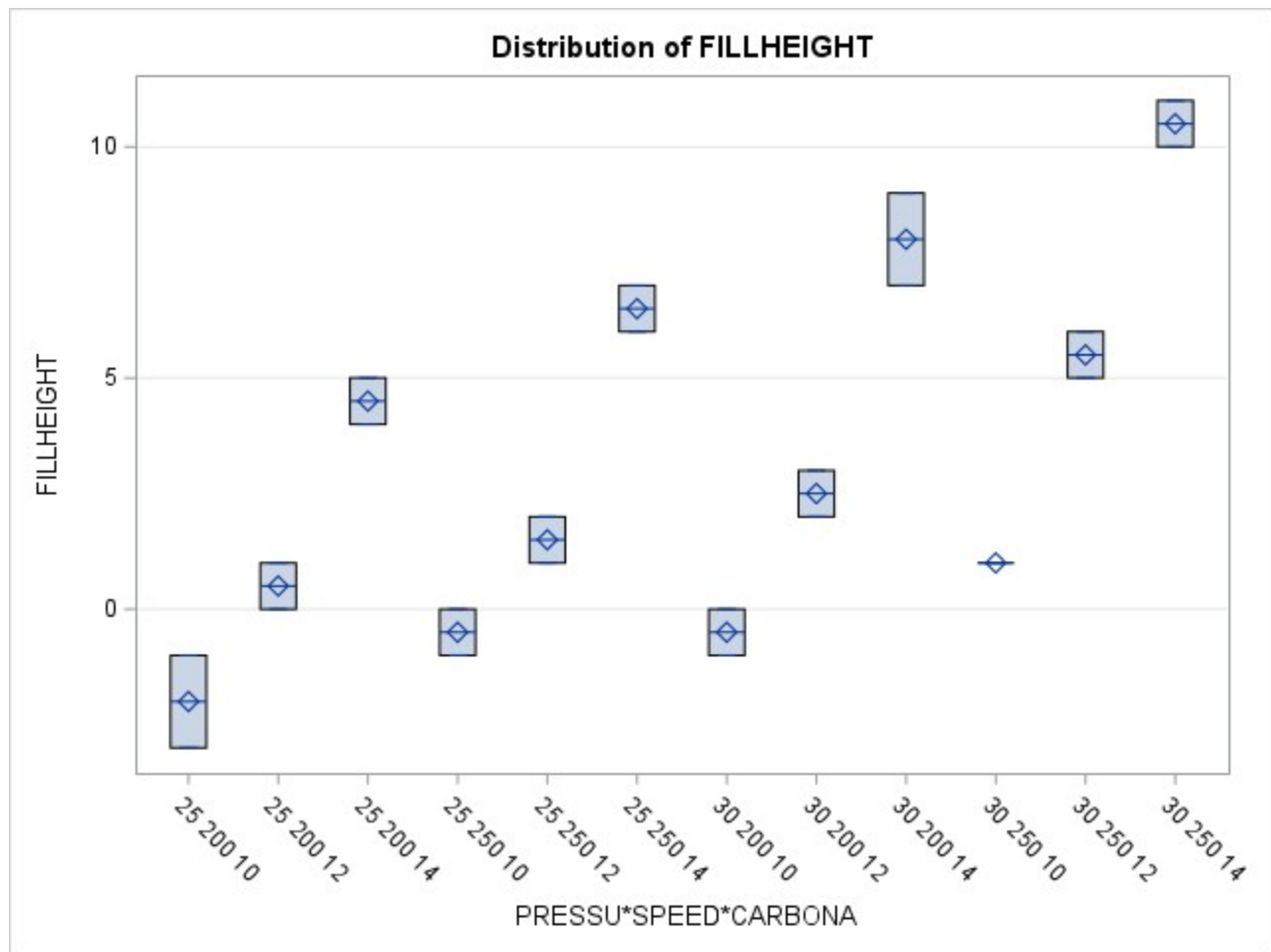
The GLM Procedure



Level of PRESSURE	Level of CARBONATION	N	FILLHEIGHT	
			Mean	Std Dev
25	10	4	-1.25000000	1.25830574
25	12	4	1.00000000	0.81649658
25	14	4	5.50000000	1.29099445
30	10	4	0.25000000	0.95742711
30	12	4	4.00000000	1.82574186
30	14	4	9.25000000	1.70782513



Level of SPEED	Level of CARBONATION	N	FILLHEIGHT	
			Mean	Std Dev
200	10	4	-1.25000000	1.25830574
200	12	4	1.50000000	1.29099445
200	14	4	6.25000000	2.21735578
250	10	4	0.25000000	0.95742711
250	12	4	3.50000000	2.38047614
250	14	4	8.50000000	2.38047614



Level of PRESSURE	Level of SPEED	Level of CARBONATION	N	FILLHEIGHT	
				Mean	Std Dev
25	200	10	2	-2.0000000	1.41421356
25	200	12	2	0.5000000	0.70710678
25	200	14	2	4.5000000	0.70710678
25	250	10	2	-0.5000000	0.70710678
25	250	12	2	1.5000000	0.70710678
25	250	14	2	6.5000000	0.70710678
30	200	10	2	-0.5000000	0.70710678
30	200	12	2	2.5000000	0.70710678
30	200	14	2	8.0000000	1.41421356
30	250	10	2	1.0000000	0.00000000
30	250	12	2	5.5000000	0.70710678
30	250	14	2	10.5000000	0.70710678

Soft Drink Bottling Experiment Example 5.3

The UNIVARIATE Procedure
Variable: RESID

Moments			
N	24	Sum Weights	24
Mean	0	Sum Observations	0
Std Deviation	0.60791876	Variance	0.36956522
Skewness	0	Kurtosis	-1.3162495
Uncorrected SS	8.5	Corrected SS	8.5
Coeff Variation	.	Std Error Mean	0.1240909

Basic Statistical Measures			
Location		Variability	
Mean	0.00000	Std Deviation	0.60792
Median	-0.00000	Variance	0.36957
Mode	-0.50000	Range	2.00000
		Interquartile Range	1.00000

Note: The mode displayed is the smallest of 2 modes with a count of 3.

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > t 	1.0000
Sign	M	-1	Pr >= M 	0.8388
Signed Rank	S	8.5	Pr >= S 	0.8137

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.863216	Pr < W	0.0039
Kolmogorov-Smirnov	D	0.252931	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.291749	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	1.631933	Pr > A-Sq	<0.0050

Quantiles (Definition 5)	
Level	Quantile

100% Max	1.0
99%	1.0
95%	1.0
90%	0.5
75% Q3	0.5
50% Median	-0.0
25% Q1	-0.5
10%	-0.5
5%	-1.0
1%	-1.0
0% Min	-1.0

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
-1.0	17	0.5	11
-1.0	1	0.5	21
-0.5	13	0.5	24
-0.5	7	1.0	18
-0.5	3	1.0	2